A craft's double hull as a ring waveguide

Gap voltage per watt of RF, DC bias, vacuum multipactor and the Kilpatrick limit, and the least RF power for a 20 kV gap — CLAUDE.md §3i

kronspice simulation report · 2026-11-07 · hull radius 5 m, height 10 m, gap 0.10 m, copper, mode 4, 1 kW

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Summary

The gap between a craft's inner and outer conducting hulls is a closed parallel-plate waveguide around the craft. Radiators in the gap drive it at one of its ring resonances, and a DC bias between the hulls is added to the RF. The questions are what gap voltage a given RF power reaches, whether the vacuum gap multipactors (a resonant secondary-electron discharge, §3i) on the way up, and what the least RF power is for a target peak voltage. This is the ring-tower experiment of examples/ring_towers moved inside a double wall.

- **1 kW gives about 1.7 kV:** at 37.685 MHz (mode 4, Q 9410) the travelling-wave gap amplitude is 1681–1683 V against the formula's 1680 V; a standing wave reaches 2379 V at its antinodes.
- **Almost every watt goes to the walls:** radiators 1000.0 W = wall loss 999.9 W + bias feed 0.141 W (travelling); 39.7 mJ of RF energy sits in the gap.
- **Unbiased ring-up multipactors:** passing 760 V the worst growth per impact is 2.02 on as-received copper (1.13 conditioned).
- **A DC bias suppresses it:** the least bias on the scanned grid that clears the whole ring-up is 500 V.
- **Far below sparking:** Kilpatrick's limit at 37.7 MHz is 8.1 MV/m, against 0.049 MV/m for the 2.5 kV-biased standing wave.
- **Evolved 20 kV design:** a 0.470 m gap at mode 1 (9.09 MHz, Q 21729) with 451 W of RF and 12496 V of bias reaches 19963 V peak, multipactor-free (worst growth 0.06).

What is being tested

Standard physics: a transmission-line resonator, the two-surface multipactor model, and Kilpatrick's empirical RF sparking limit. A DC bias across a coaxial gap is the measured cure for multipacting in the LEP2 352 MHz couplers (2.5 kV) and the KEKB 509 MHz couplers (2 kV); the example asks whether the same works for a hull-sized gap. The same multipactor model drives the electron pumps of examples/photon_electron.

The model

- **The gap as a line.** The parallel-plate TEM line of the telegrapher's equations (Paul §4.2): per unit length around the hull $L' = \mu_0 \cdot d/w$, $C' = \varepsilon_0 \cdot w/d$ and $Z_0 = \eta_0 \cdot d/w$, with d the gap and w the hull height. Wall loss $R' = 2 \cdot R_s/w$ with $R_s = \sqrt{(\pi \cdot f \cdot \mu_0 \cdot \rho)}$ the skin-effect surface resistance of each copper wall, so $\alpha = R_s/(\eta_0 \cdot d)$. N sections (24 per wavelength) close the ring; here 96 sections and $Z_0 = 3.77\ \Omega$.
- **Solve.** The AC analysis gives the steady state at the drive frequency, the operating point gives the bias, and the gap voltage is their sum. Expected travelling-wave amplitude at mode n: $\sqrt{(Z_0 \cdot P_{in} \cdot Q/(\pi \cdot n))}$; a standing wave doubles it at the antinodes.
- **Bias.** The DC bias charges the outer hull through a high resistance, stores $\frac{1}{2} \cdot C_{gap} \cdot V_{dc}^2$ once and draws no steady DC current. The gap then swings between $V_{dc} \pm V_{rf}$.
- **Multipactor check** (em/multipactor.go; Hatch & Williams 1954, Vaughan 1988). Electrons launched from either wall at every RF phase with 2 eV are tracked through the combined RF + DC field impact after impact, multiplying Furman's yield $\delta = \delta_{max} \cdot s \cdot x/(s - 1 + x^s)$ at each one, with measured copper values (Baglin et al. 2002: as received δ_{max} 2.03 at 262 eV, s 1.39; conditioned 1.13 at 318 eV, s 1.35). A phase whose mean yield per impact stays above 1 is a growing discharge. Ring-up passes through every amplitude below the final one,

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so each is checked.

- **Kilpatrick's limit** f[MHz] = 1.64·E²·exp(−8.5/E), E in MV/m, bounds the peak gap field.

Results

Example hull: radius 5 m, height 10 m, gap 0.10 m, copper, mode 4, 1 kW

All four runs resonate at f = 37.685 MHz with Q = 9410 and Z_0 = 3.77 Ω.

run	RF gap amplitude	formula (travelling)	peak gap (RF + bias)	wall loss	bias feed	bias energy
travelling, 0 V	1681–1683 V	1680 V	1683 V	999.9 W	0.141 W	0 mJ
travelling, 2500 V	1681–1683 V	1680 V	4183 V	999.9 W	0.141 W	87.794 mJ
standing, 0 V	0–2379 V	1680 V	2379 V	999.7 W	0.283 W	0 mJ
standing, 2500 V	0–2379 V	1680 V	4879 V	999.7 W	0.283 W	87.794 mJ

In every run the radiators deliver 1000.0 W, which the wall loss and the bias feed take, and 39.7 mJ of RF energy is held in the gap (Q·P/ω). The 2.5 kV bias stores its 87.794 mJ once.

Multipactor in the vacuum gap (f·d = 3.77 MHz·m)

Worst-phase mean secondary yield per impact, as-received copper, with the impact energy; values above 1 (marked susceptible) grow.

V_{rf}	bias 0 V	bias 500 V	bias 2500 V	bias 5000 V
100 V	0.37 at 14 eV	0.07 at 25 eV	0.06 at 126 eV	0.06 at 248 eV
300 V	**1.10 at 53 eV, susceptible**	0.07 at 22 eV	0.06 at 115 eV	0.06 at 238 eV
1000 V	**1.78 at 630 eV, susceptible**	0.11 at 177 eV	0.07 at 88 eV	0.06 at 206 eV
1700 V	0.29 at 732 eV	0.16 at 398 eV	0.07 at 60 eV	0.06 at 171 eV
3400 V	0.08 at 341 eV	0.08 at 334 eV	0.08 at 14 eV	0.06 at 113 eV
10000 V	0.06 at 831 eV	0.06 at 835 eV	0.07 at 376 eV	0.07 at 673 eV

Ring-up

copper	ring-up to	worst growth, no bias	least bias that clears it
as received	1680 V (travelling)	2.02 at 760 V	500 V
as received	2375 V (standing antinode)	2.02 at 759 V	500 V
conditioned	1680 V (travelling)	1.13 at 760 V	500 V
conditioned	2375 V (standing antinode)	1.13 at 759 V	500 V

Kilpatrick's RF limit at 37.7 MHz is 8.1 MV/m; the 2.5 kV-biased standing wave peaks at 0.049 MV/m.

Evolved design

The GA chooses gap, mode, RF power and bias for a 20000 V peak, at least 5000 V of it RF, with the least RF power, a multipactor-free ring-up and a field under Kilpatrick's limit.

quantity	best of 332 evaluations
gap, mode	0.470 m, mode 1 (9.09 MHz, Q 21729)
RF power, bias	451 W, 12496 V
peak gap voltage	RF 7467 V + bias 12496 V = 19963 V
multipactor	ring-up worst growth 0.06 (at 7247 V)
field	0.042 MV/m against Kilpatrick's 5.3 MV/m
ledger	radiators 451.0 W = wall loss 448.3 W + bias feed 2.720 W
bias supply	delivers C·V² = 0.97 J once: 0.48 J stored, 0.48 J lost in the feed resistor

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Ledger and validation

- Power balance in every run: radiator input = wall loss + bias feed; no steady DC current (no gap leakage is modelled).
- Tests (main_test.go) hold the resonance to $n \cdot c/\ell$ times the ladder's lattice factor to 10^{-9}, the amplitude to the formula within 1 %, Furman's yield, the multipactor zones and bias suppression, and Kilpatrick's inverse.
- Kron twin: the exported 193-node circuit, torn into 6 pieces, equals MNA to 5.6×10^{-9} with the same power.

Limits of the model

- Space charge and the spread of emission energies are left out of the multipactor check, so it marks where a discharge can start, not how large it grows.
- Gap leakage is not modelled, so the bias draws no steady current.
- The bias scan is on a grid (0, 500, 2500, 5000 V): 500 V is the least grid value that clears ring-up, not a threshold.
- The hull is an ideal parallel-plate ring line with copper walls; real hull penetrations and supports are not modelled.

Reproducing the results

```
go run ./examples/double_hull
go test ./examples/double_hull
python examples/double_hull/viz/make_report.py examples/double_hull/double_hull_report.pdf
```

Sources: Paul, Analysis of Multiconductor Transmission Lines, 2nd ed., §4.2; Hatch & Williams (1954); Vaughan, IEEE Trans. Electron Devices 35, 1172 (1988); Baglin et al., CERN LHC Project Report 472 (2002); Kindermann et al., EPAC 1996; Kijima et al., EPAC 2000; Kilpatrick, Rev. Sci. Instrum. 28, 824 (1957); Boyd, LANL LA-UR-82-1166 (1982).

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